

Building Energy

Sector	Intervention detail	Intervention (short)
What-if sprawl	What if there is no Met Council plan. 2040 has the same single-family proportion as 2018.	Met Council multi-family housing effects
Res	Compact buildings : half of new SF homes become MF	Compact buildings (50% MF new)
Res	Apply 0.8% reduction in Electricity EUI for 1% tree canopy change, scale accordingly	Tree canopy energy reduction
Res	Affordable FA; All SF grow only by 5% vs now at 15%	Affordable floor area
Res	New home efficiency; 50% new homes to LEED Gold (25 kBtu/sf; electricity reduces by 64% and gas by 64%)	New homes (50%) are LEED-Gold
Res	Existing home: 80% reduced EUI by 33%	Existing homes (80%) retrofitted
Res	Passive Buildings for remaining 20% stock at 66% reduction	Existing homes (20%), passive homes
Comm	80% of existing commercial buildings are high-efficiency (reduce EUI by 25%)	Existing commercial bldgs (80%), high efficiency
Res	All homes receive effective messages, use in-home display, and smart meters (reduce HH energy use by 11%)	Behavioral & smart nudges (100% HH)
Comm	Smart grid and technologies reduce commercial electricity use by 12% ()	Smart grid efficiency (100% comm buldgs)
Ind	Smart grid and technologies reduce industrial electricity use by 12% ()	Smart grid efficiency (100% ind)
Grid	Grid decarbonization: GHGs reduced 80% (compared to 2018)	Grid decarbonization, 113 kgCO ₂ e/MWh
Heating transition	59% of homes use electric-based heat pumps and water heaters	Electrifying residential heating (59% minus 14% HH)
Heating transition	59% of commercial buildings (up from 19%) use electric-based heat pumps and water heaters)	Electrifying commercial heating (59% minus 19%)
Renewable natural gas	Renewable natural gas (RNG) displaces remaining fossil natural gas demand	Renewable natural gas (RNG) displaces remaining fossil natural gas demand

Greenhouse Gas Scenario Planning Tool

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Glossary

- **Single-family housing:** an independent residential structure that sits on its own land and is designed to be used as a single dwelling unit, without shared walls or utilities.
- **Multi-family housing:** A building, or a complex with several buildings, that contains multiple separate housing units for residential inhabitants.
- **Compact buildings:** buildings designed to reduce sprawl and use land and resources more efficiently
- **Home retrofits:** Renovations or equipment installed in existing homes to increase their energy efficiency
- **EUI:** Energy use intensity; expresses a building's energy use as a function of its size or other characteristics
- **Affordable floor area:** amount of floor area in an affordable housing unit
- **LEED:** Leadership in Energy and Environmental Design; the most widely used green building rating system in the world.
- **Passive Buildings:** structures adhering to a highly energy efficient building standard that also promotes indoor comfort and acoustic insulation.
- **Smart Meters:** an electronic device that records information such as consumption of electric energy, voltage levels, current, and power factor and communicates the information to the consumer and electricity suppliers.
- **Electric-based heat pumps:** An air source heat pump that uses a small amount of energy to transfer a large amount of heat, without emitting greenhouse gases.
- **Renewable Natural Gas:** A type of fuel that is produced through the decomposition of organic matter and processed to purity standards that match that of conventional natural gas.